

Data Ethics Checklist

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Internal verification for ethical data sourcing and handling. Score: 23/30 items complete (77%).
Remaining 7 items pending first commercial deployment.

Section 1 — Data Collection & Sourcing

#	Check	Status	Notes
1.1	All data sources have verified open licenses	✓ Complete	Sentinel-1, CHIRPS, SRTM, SoilGrids, ERA5, GPM, OSM — all open
1.2	No personal data collected	✓ Complete	Addresses pseudonymized (SHA-256) before storage
1.3	No demographic proxies used as features	✓ Complete	Only geophysical variables in model
1.4	Data sources documented in register	✓ Complete	See AI Inventory List
1.5	Institutional data used only under formal agreement	⌚ Pending	ANACIM, ONAS partnerships in negotiation
1.6	Training labels documented and auditable	✓ Complete	SAR-derived + terrain sources cited

Section 2 — Data Storage & Security

#	Check	Status	Notes
2.1	Data hosted in sovereign datacenter	✓ Complete	Sénégal Numérique SA, Diamniadio (ISO 27001)
2.2	Data encrypted at rest	✓ Complete	SHA-256 for addresses
2.3	Data encrypted in transit	✓ Complete	TLS on all API endpoints
2.4	Access controls documented	✓ Complete	CTO-only access to partner data
2.5	Retention policy defined	✓ Complete	API logs: 90 days, scores: 24 months
2.6	No cross-border data transfer without consent	✓ Complete	West Africa only

Section 3 — Fairness & Bias

#	Check	Status	Notes
3.1	Model trained on geophysical data only	✓ Complete	No socioeconomic variables

#	Check	Status	Notes
3.2	Fairness audit scheduled quarterly	 Pending	First audit due Q3 2026
3.3	Score distribution reviewed across zones	 Pending	Pending first commercial deployment
3.4	No systematic over-scoring of low-income zones	 Pending	To be verified at first audit
3.5	Bias audit results published for clients	 Pending	Protocol defined, implementation pending

Section 4 — Model Transparency

#	Check	Status	Notes
4.1	Model Card published	 Complete	Available on request
4.2	Feature importance documented	 Complete	XGBoost: sar_delta 50%, precip 15%
4.3	Known limitations documented	 Complete	SRTM resolution, 40-sample dataset
4.4	Performance metrics published	 Complete	Recall 83.3%, Precision 89.3%, F1 86.2%
4.5	Model version tracked per API response	 Complete	model_version field in every response

Section 5 — Human Oversight

#	Check	Status	Notes
5.1	Mandatory human review for scores > 75	 Complete	Enforced at API level
5.2	No automated insurance approvals/denials	 Complete	API returns decision support only
5.3	Human override mechanism exists	 Complete	POST /api/v1/feedback
5.4	Override decisions logged	 Pending	Implementation pending
5.5	Responsible officer designated	 Complete	Babacar Ndao, CTO

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23/30 items complete (77%). All critical safety and data sovereignty items are complete. Remaining items are operational and will be completed at first commercial deployment.